

OVERVIEW

Real-world lessons often lead to big open-ended questions -- ones in which students may draw on mathematical analysis as well as experiences, beliefs, and opinions to construct responses that are entirely their own. While there's not a single right answer, some are clearly stronger than others. In this conversation, you'll explore how to make sense of and strengthen student responses to real-world questions that do not have a single right answer.

SPARK MEDIA & FOCUS LESSON

Student Work

Sample student responses to an open-ended question, in the lesson [Coupon Clipping](#).

PEDAGOGICAL THEME

Focusing on the Real World

USING THE CONVO GUIDE FOR A GREAT ROUNDTABLE DISCUSSION

- This conversation is designed for a group of 2-5 educators over 30-60 minutes.
- The Convo Guide serves as an agenda for a roundtable conversation where everyone is there both to learn and to share their expertise.
- Each participant should have their own copy of the Convo Guide and access to the Spark Media.
- Pace yourselves to get through the entire agenda in the time you have.
- Questions aren't intended to have a single right answer; instead, they're designed to spark rich discussions.
- At the end, everyone will commit to trying "one small thing" the very next day. Consider how you might follow up with one another!

OBSERVE

5-15 MIN

- Strong responses to open questions typically **make a clear claim, include evidence and/or reasoning, acknowledge the counterargument, and consider the audience or perspective.** Pick one of the two pages in the Spark Media to review.
- What are some strengths and weaknesses in each response?
- What can be *challenging* for students about these open questions? What can be *inviting*? Consider students from across the spectrums of achievement and readiness in your math class.

DISCUSS

10-15 MIN

- What follow-up questions could a teacher use in the moment to nudge students to make one of these weak responses stronger?
- As a teacher, how should your approach to “open” questions be different than it is for questions with one right answer? What should be the same?

RELATE

10-15 MIN

- What skills and mindsets do students need to engage with questions without right answers in math class?
- What activities, practices, etc. do you already do to build and reinforce those skills and mindsets over time? What new things could you try?

...AND COMMIT!

5-10 MIN

- What is one small thing you can do tomorrow in support of those shifts?

SAMPLE STUDENT RESPONSES, POSITION “FOR”

The following are sample written responses from students during the Citizen Math lesson Coupon Clipping in which students reason with percents and proportions to evaluate enticing coupons and debate whether retailers should be allowed to raise the price of items in order to then put them on sale. All responses take the same stance, but offer different levels of support.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

I think businesses can decide the price however they want. If you don't like it, go shop somewhere else.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

Well there's a lot of ways to do sales. Some companies raise prices first and others might already have higher prices before they put things on sale.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

If a business wants to make fake sales, I think that's fine because you don't have to shop there. My reason is that people like sales even if the price is higher. So it's just smart business to mark things as "on sale" even if you're charging the same price you were always going to charge

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

Stores should be able to make sales however they want!

My evidence that customers want sales is that when JC Penney tried to stop having "fake" sales everyone stopped shopping there. So they had to bring back the sales to make customers happy and want to shop there again.

Someone might counter that a bunch of stores were sued for cheating customers with "fake" discounts. But from the customer's perspective those BIG sales are still what they want so no one is really harmed. Plus, there is no "real" price for items in the first place because the price is whatever people are willing to pay.

SAMPLE STUDENT RESPONSES, POSITION “AGAINST”

The following are sample written responses from students during the Citizen Math lesson Coupon Clipping, in which students reason with percents and proportions to evaluate enticing coupons and debate whether retailers should be allowed to raise the price of items in order to then put them on sale. All responses take the same stance, but offer different levels of support.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

I think businesses should not be able to lie about prices. That's not fair to customers!

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

They are deceiving customers when they publicize a big discount on a fake price that they'd never actually charge.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

Even though the store isn't actually stealing money from the customer, they are still tricking people into thinking their getting more value than they actually are.

Q5b. Do you think stores should be allowed to raise prices before offering discounts? Explain.

Stores should not be allowed to raise prices just for sales because it's dishonest.

My evidence is that the stores like JC Penney are increasing prices just so they can trick people into thinking they are getting big discounts.

Even though JC Penney found out their customers wanted the fake sales, it is still misleading people into thinking they are buying something of a higher value than it actually is. So it's fraud and should not be allowed.